

Real-Time Video Processing Web-App

Link to our GitHub Repository

Link to the User Stories discussed so far



[GitHub Repo](#)



[User Stories](#)

PS:

Develop a web application that will perform real-time video processing of a webcam video stream consisting of a green screen, replacing it with a desired video or image.

Ideation:

Once the user uploads the video clip or Image, the user should be able to use it as a background provided that he/she has a green screen behind. The video processing will be done from scratch and using [Chroma Keying](#) technique we will change the green screen into the desired background.

Approach:

Designing of the Web-App is done with the help of Html, CSS and Javascript. For font styling, we used Google-fonts API. The Homepage of the webpage gives the user the option to go directly to the page where he/she can upload the video or image as a background and start streaming the video which will be processed in real-time. Also, Bootstrap-5 is used to make some pages more attractive. The whole website is responsive and can be used with different screen sizes. The about us page provides a way to reach out to the developers of the Web App directly.

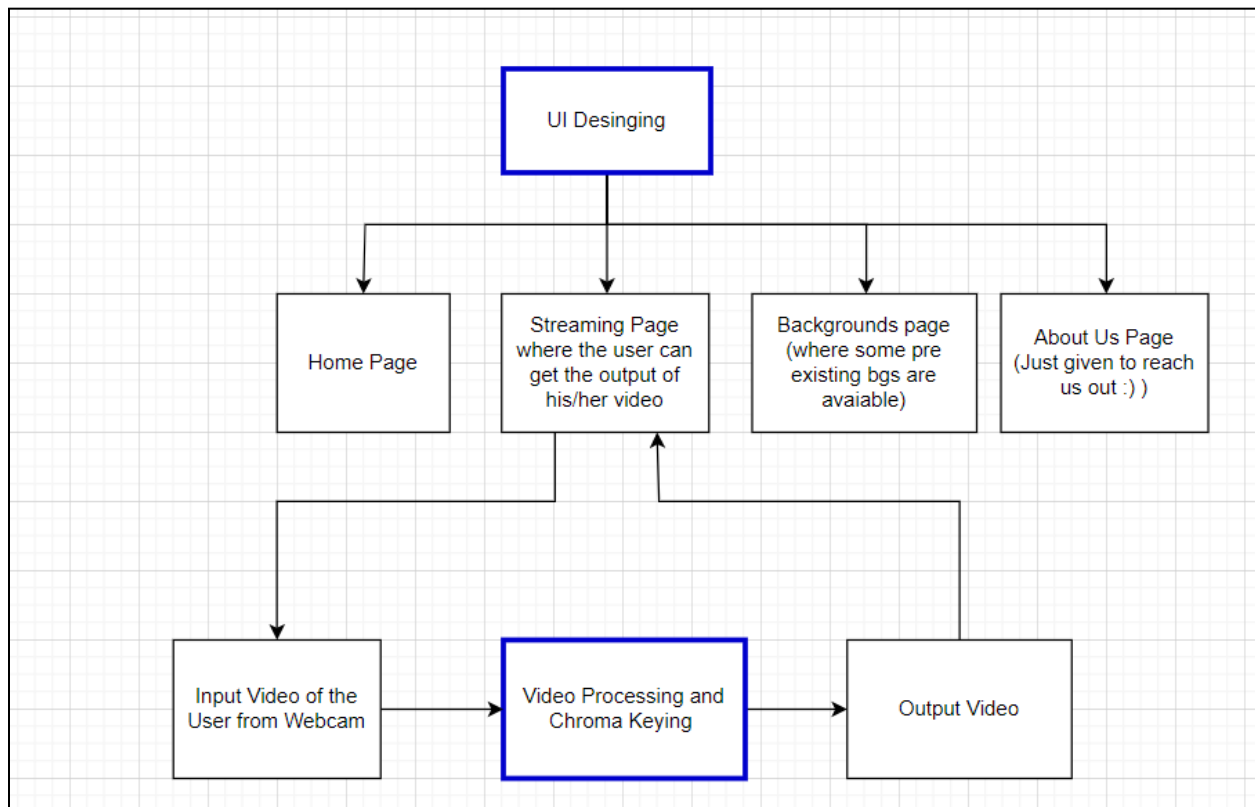
Tech-Stack:

1. HTML
2. CSS
3. Javascript
4. Bootstrap

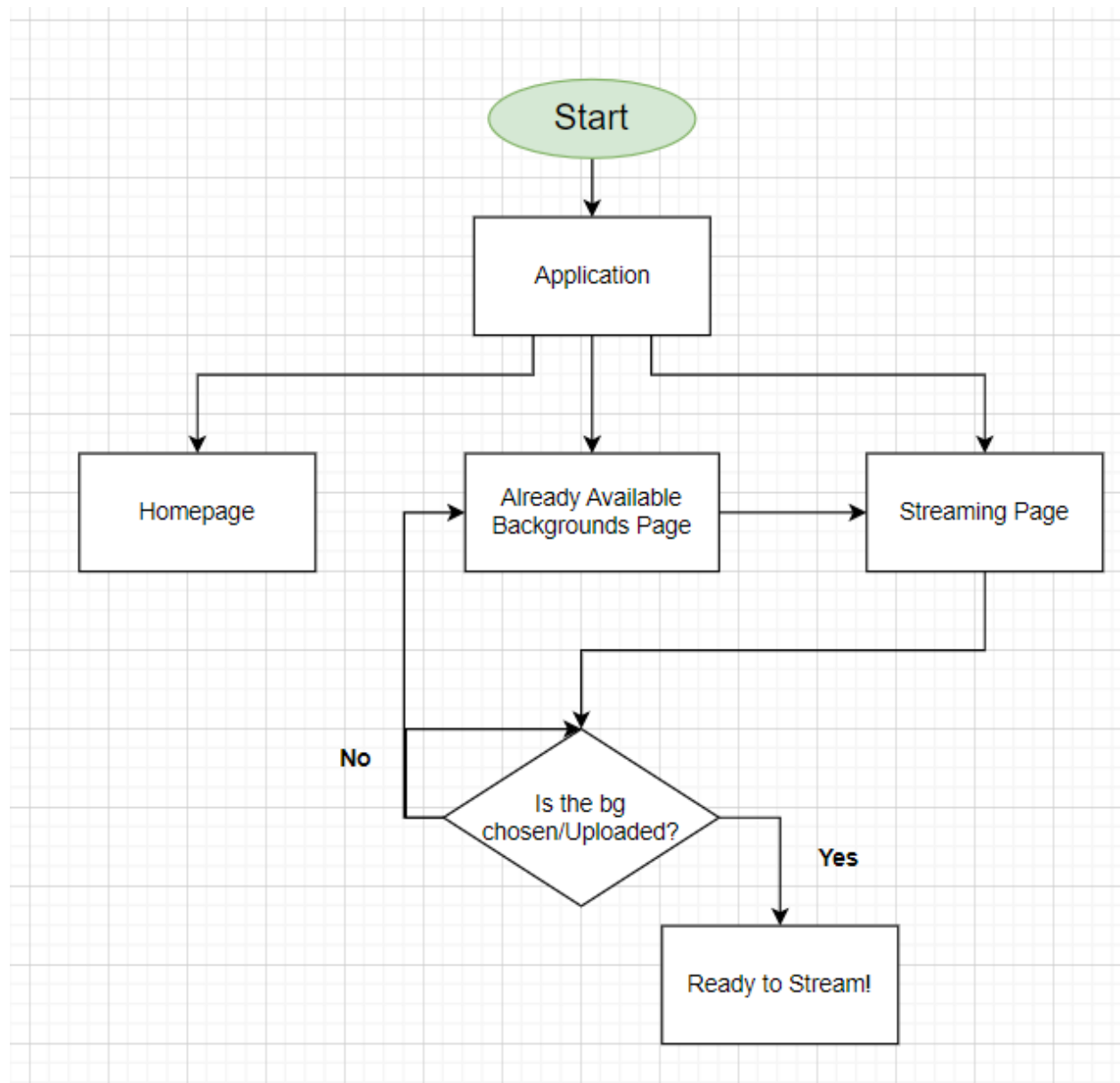
Dependencies:

1. Google-fonts API
2. Bootstrap v5.0.2

Flow Chart of Work:



Userflow Diagram



Contributions:

Anadi Sharma (B21CS008)

Using Figma, Html, CSS and Javascript designed and made the homepage and the backgrounds page.

Connected the Video Processing and Chroma Keying part of the project to the streaming page and backgrounds page.

Made all the pages responsive to different screen sizes.

Piyush Jhamnani (B21ME046)

With the help of Javascript taking input through a web-cam and process real-time background subtraction which gives the utility to work on both images as well as with video in the background

Anushkaa Ambuj (B21ES006)

&

Mayank Jonwal (B21AI022)

Worked on accessing the webcam on the Html page for video streaming using getUserMedia API through javascript.

Kushal Agarwal(B21BB017)

Improved the working of the chroma key,

Earlier, frame extraction and output was done on the same canvas resulting in significant lag, I separated the process of frame extraction and output into two different canvasses.

Samarth Sudhirkumar Bhalerao (B21EE060)

Designed the streaming and feedback pages for the web application using HTML, CSS and JavaScript.

Provided real-time unprocessed webcam output to the main streaming page.

Provided an identifying icon to web pages.

